Ho An Loc

0766552398 — th0935057511@gmail.com — https://github.com/hal2332004

Education

FPT University **2022 – 2026**

- Bachelor of Artificial Intelligence
- GPA: 3.24/4.0

Professional Summary

AI Engineer with hands-on experience in Computer Vision, NLP, OCR, and LLM-powered applications, including chatbots built with RAG and GraphRAG pipelines and memory-enabled conversational systems. Skilled in developing end-to-end AI solutions, optimizing model efficiency through knowledge distillation and feature engineering, and quantizing and deploying models on edge devices. Experienced in designing and developing full-stack applications and automating workflows with n8n to integrate AI models, backend services, frontend applications, and third-party APIs. Focused on delivering efficient, scalable, and production-ready AI solutions.

Research Experience

Enhanced Plant Leaf Disease Classification Using Machine Learning and Synthesized Features

Computer Vision • Feature Engineering • Image Classification

- Developed a feature extraction framework combining handcrafted color, texture, and shape descriptors with SIFT-BOW features for plant leaf disease classification.
- Reduced feature dimensionality by approximately **78.2%**, from 316 to 69 features, using Random Forest-based feature selection to improve computational efficiency.
- Evaluated six machine learning classifiers and achieved **98.51% accuracy** using SVM with 10-fold cross-validation on the PlantVillage dataset.
- **GitHub:** Plant Leaf Diseases Classification

Efficient Rice Seed Classification Using Knowledge Distillation via ResNet18

Deep Learning • Computer Vision • Model Distillation

- Developed **ResNet18-Lite**, a lightweight rice seed classification architecture integrating Knowledge Distillation, Depthwise Separable Convolutions, and CBAM attention mechanisms.
- Reduced trainable parameters from **11.18M to 0.82M (92.64% reduction)** compared with the original ResNet18, reducing model complexity while maintaining competitive classification performance.
- Achieved **97.78% average classification accuracy**, outperforming the baseline ResNet18 at 96.84% across six rice varieties using a combined loss strategy.

Work Experience

FPT Software – AI Engineer Intern **January 2025 – May 2025**

Computer Vision • OCR • Document AI

- **Deed (OCR System):** Developed an intelligent document information extraction system for legal and administrative documents, including land certificates, housing records, and tax forms.
- Built the document understanding pipeline covering image preprocessing, OCR, key-value extraction, and structured output generation for multiple document templates.
- Optimized preprocessing techniques to improve OCR robustness on low-quality scanned documents.
- Collaborated with backend engineers to deploy AI inference services through RESTful APIs.
- **Tech Stack:** Python, PyTorch, PaddleOCR, OpenCV, FastAPI, Docker, Git.

FlappieDoors – AI Engineer (Remote) **March 2025 – May 2025**

Computer Vision • Image Classification • Edge AI

- **Lens Condition Detection for Smart Pet Doors:** Developed a deep learning system to detect camera lens conditions in Flappie smart pet doors, classifying fisheye camera images into three categories: clean, dirty, and occlusion.
- Designed an image classification pipeline for grayscale and infrared nighttime images, incorporating preprocessing, CLAHE enhancement, data augmentation, and automated labeling to improve data quality.
- Evaluated multiple CNN architectures and achieved **98.14% test accuracy** with MobileNetV2 on a nighttime test set of 215 images.
- Developed a model conversion pipeline using TensorFlow/Keras and ONNX, preparing the model for quantization and deployment on the **Rockchip RK3566 NPU**.
- **Tech Stack:** Python, TensorFlow/Keras, MobileNetV2, OpenCV, Albumentations, scikit-image, Weights & Biases, TensorBoard, ONNX, ONNX Runtime, RKNN Toolkit2, Docker.

AI Automation and Chatbot Development **Aug 2025 – May 2026**

- **Workflow Automation:** Designed and developed automated workflows using n8n to streamline business processes, automate repetitive tasks, and orchestrate multi-step operations.

- **Third-Party API Integration:** Integrated external REST APIs to connect business platforms and backend services, enabling automated data exchange and cross-platform workflows.
- **Customer Service Chatbots:** Designed and developed customer consultation chatbots for businesses on social media and messaging platforms to automate customer interactions and handle inquiries.
- **System Integration:** Connected social media platforms, external APIs, and business services to build end-to-end automation solutions tailored to operational requirements.
- **Tech Stack:** n8n, REST APIs, Webhooks, JSON, JavaScript, Python, Social Media APIs, Supabase, RAG.

Software Engineer (Remote) **Aug 2026 – Oct 2026**

- **Web Development:** Developed an internal management system and designed responsive landing pages to support business operations and improve online user experience.
- **System Implementation:** Built user interfaces with React, TypeScript, Vite, and Tailwind CSS. Integrated Supabase for authentication, database management, and real-time data synchronization, with role-based access control and Row-Level Security.
- **UI/UX Design:** Designed responsive, user-friendly interfaces with a focus on visual consistency, usability, and cross-device compatibility.
- **Deployment and Maintenance:** Deployed web applications to a Linux VPS using Nginx, managed code through Git and GitHub, and documented deployment and maintenance procedures.
- **Tech Stack:** React, TypeScript, Vite, Tailwind CSS, Supabase, PostgreSQL, Git, GitHub, Linux, Nginx.

Selected AI Projects

Vietnamese Speech Chatbot **May 2025 – August 2025**

LLM • RAG • Speech AI

- Developed an end-to-end Vietnamese voice chatbot capable of understanding spoken input and generating natural spoken responses.
- Designed a Retrieval-Augmented Generation (RAG) pipeline integrating Automatic Speech Recognition (ASR), vector retrieval, Large Language Models, and Text-to-Speech (TTS).
- Ran the LLM locally on CPU using llama.cpp, achieving responsive inference for voice conversations, and built asynchronous RESTful APIs with FastAPI for the web interface.
- Containerized the complete application using Docker for simplified deployment and portability.
- **Tech Stack:** FastAPI, Docker, Whisper, Gemini API, LangChain, Qdrant, llama.cpp, XTTS, Sentence-Transformers, HTML/CSS, JavaScript, Git.

Real-Time Traffic Video Super-Resolution on Edge Devices **December 2025 – May 2026**

- **Dataset Construction:** Collected and preprocessed traffic videos from Da Nang, building a dataset of 5,857 low-resolution/high-resolution image pairs through a degradation simulation pipeline that reproduces real-world webcam noise and compression artifacts.
- **Model Training and Evaluation:** Trained and fine-tuned 8 Super-Resolution models, comparing image quality using PSNR, SSIM, and LPIPS, alongside inference latency and memory consumption.
- **Edge Optimization:** Optimized models for NVIDIA Jetson Orin NX using TensorRT and FP16/INT8 quantization, analyzing trade-offs between image reconstruction quality and inference performance.
- **System Integration and Deployment:** Integrated the selected model into a Docker-based application supporting single-image enhancement, offline video processing, and real-time webcam streaming via RTSP. Tested and monitored system performance under practical deployment conditions.
- **Tech Stack:** Python, PyTorch, OpenCV, Super-Resolution, TensorRT, FP16/INT8 Quantization, NVIDIA Jetson Orin NX, Docker, Git.

Technical Skills

- **Programming Languages:** Python, SQL, C++, R
- **AI & Machine Learning:** PyTorch, TensorFlow, Scikit-learn, LLM APIs, LangChain, Sentence-Transformers, HuggingFace
- **Data Science & Computer Vision:** Pandas, NumPy, OpenCV, Matplotlib, Seaborn
- **Web Development:** React, Node.js
- **Backend & MLOps:** FastAPI, Docker, Git/GitLab, Ollama, vLLM, llama.cpp
- **Databases & Vector DB:** Qdrant, MySQL, MongoDB, Redis, Supabase

Language Skills

- **English:** Intermediate.
- **Vietnamese:** Native.